

Soumya Sri Kesani

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EDUCATION

University of Oklahoma, Norman, OK	Aug 2024 – May 2026
M.S. in Data Science and Analytics	GPA: 3.90 / 4.00
CVR College of Engineering, Hyderabad, India	Graduated May 2024
B.Tech in Computer Science Engineering	GPA: 3.07 / 4.00

EXPERIENCE

Graduate Research Practicum Jan 2026 – May 2026

Data Institute for Societal Challenges (DISC), University of Oklahoma | [Live Demo](#)

- Engineered a **spatiotemporal clustering pipeline** in R processing **1.85 million NASA VIIRS SNPP thermal detections** (2013–2024) across 53 African countries to detect wildfire encroachment in the East-West Guinean Lowland Forests ecoregion; sponsored by Dr. Michael Wimberly (DISC).
- Implemented **graph-theoretic clustering** using igraph and Delaunay triangulation with optimized spatial ($D = 2,000$ m) and temporal ($T = 2$ days) thresholds to reconstruct **780,000+ discrete fire events** from raw hotspot detections.
- Performed **IDW spatial interpolation** (gstat) and burn date rasterization (terra) to generate annual 375 m resolution raster outputs; validated against Landsat dNBR masks via Google Earth Engine.
- Achieved **mean Kappa of 0.588** across six forest reserve validation sites; pipeline detected **14.1% more burned area** than the MODIS MCD64A1 global standard; delivered a documented R codebase on GitHub to the project sponsor.

IT Support and Service Operations May 2025 – May 2026

University of Oklahoma, Norman, OK

- Managed security groups, access controls, telecom workflows, and Tier-1 technical support across campus systems; improved ticket resolution through structured workflow documentation.

Research Assistant Jan 2023 – May 2023

CVR College of Engineering, Hyderabad, India

- Co-authored and published research on Geographic Alert Systems in IJCERT. [View Paper](#)

TECHNICAL SKILLS

Languages	Python (NumPy, Pandas, Scikit-learn, SHAP, pdfplumber), R (sf, terra, igraph, gstat, ggplot2), SQL
Data Eng.	ETL/ELT Pipeline Design, PDF Data Extraction, Large-scale Ingestion, REST API Integration, Schema Design, Stored Procedures
Databases	MySQL, Snowflake, Microsoft Azure SQL, SQLite
Cloud / Tools	Microsoft Azure, Snowflake, Git, GitHub, Django, Streamlit, Google Earth Engine, Claude Code
ML / NLP	Random Forest, Logistic Regression, Naive Bayes, ARIMA, ETS, K-Means, Isolation Forest, SHAP Explainability, TF-IDF Vectorization, NLP Text Classification
Visualization	Tableau, Power BI (DAX), Excel (Pivot Tables, Dashboards), Matplotlib, ggplot2

PROJECTS

NLP-Based Customer Churn Prediction Pipeline *Python, Snowflake, Streamlit, TF-IDF, Scikit-learn*

- Built a **cloud-based NLP pipeline** processing **767,000+ CFPB consumer complaints**; designed an end-to-end data flow from raw CSV ingestion into Snowflake to a live Streamlit triage dashboard for churn risk prediction.
- Applied **TF-IDF vectorization** and trained Logistic Regression, Naive Bayes, and Random Forest classifiers on combined structured and text features; evaluated using F1-score, precision, and recall.

Automated Bank Statement ETL System with ML Classification *Python, pdfplumber, Pandas, Scikit-learn*

- Engineered a **multi-format PDF extraction pipeline** using pdfplumber and Pandas that extracts, normalizes, and validates transactions into a schema-consistent CSV output, serving as the data layer for a full-stack finance dashboard.
- Implemented **ML-based transaction categorization**, spending forecasting, and Isolation Forest anomaly detection on extracted financial records.

Heart Disease Risk Stratification: EarlyMed Dataset *Python, Pandas, Scikit-learn, SHAP, Matplotlib, Excel*

- Processed a **70,000-row, 19-feature clinical dataset**; engineered features including WHO BMI risk bands and stress-sleep interaction terms; trained Random Forest and Logistic Regression classifiers with **AUC 0.91**.
- Applied **SHAP explainability** to identify blood pressure, cholesterol, and family history as top predictors; delivered an interactive Excel pivot dashboard for non-technical clinical stakeholders.

National Parks Service Database System *SQL, MySQL, Microsoft Azure*

- Designed ER diagrams and a **3NF-normalized relational schema**; implemented in MySQL with primary/foreign key constraints, indexed columns, and stored procedures; deployed to **Microsoft Azure** with role-based access controls and validated end-to-end query performance.

US Natural Gas Consumption Forecasting (1973–2025) *Python, Pandas, ARIMA, ETS, Scikit-learn*

- Built an **end-to-end time-series pipeline** on 627 months of EIA data: STL decomposition, ADF stationarity testing, and rolling-window model comparison across baseline, regression, ETS, and ARIMA specifications using MAE, RMSE, and MAPE.

Stock Market Interactive Dashboard *Tableau, Excel*

- Cleaned and transformed multi-year stock market datasets spanning approximately 2.5 million records; built interactive **Tableau dashboards** with time-series trend lines, volume overlays, and sector performance indicators with drill-down filtering.

Energy Consumption Forecasting: Steel Manufacturing *Python, Scikit-learn, K-Means, Isolation Forest, Regression*

- Processed **35,040 industrial sensor records**; built regression forecasting models, applied **K-Means clustering** (silhouette-scored) to segment plant operations into **3 distinct operational regimes** (peak load, idle, transition).
- Implemented **Isolation Forest anomaly detection** to flag unusual energy consumption spikes indicating potential equipment faults, supporting proactive maintenance decisions.

CERTIFICATIONS

Data Analytics Job Simulation	<i>Deloitte Australia / Forage</i> : Simulated real-world data analytics tasks including data cleaning, dashboard creation, and deriving business insights.
Claude Code 101	<i>Anthropic / Skilljar</i> : Agentic coding workflows, CLI-based code generation, and responsible LLM tool use in professional development environments.
Prompt Engineering with Chat-GPT	<i>LinkedIn Learning</i> : Structured prompting including chain-of-thought, role assignment, and iterative refinement for data analysis workflows.
Hands-On SQL	<i>LinkedIn Learning</i> : Practical SQL covering joins, subqueries, window functions, and query optimization on real-world relational datasets.

SUMMARY

Data professional with hands-on experience in data engineering, machine learning, NLP, and geospatial analytics. Passionate about building scalable pipelines, automating data workflows, and delivering insights through visualization and predictive modeling. Committed to leveraging AI-augmented tools to accelerate analysis and surface actionable results from complex, large-scale datasets.